

mitochondria are a common thread that unites all eukaryotes.

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Vision: More Than Expected in the Early Visual System

In the standard model of central visual processing, orientation tuned responses in cortex are built from untuned thalamic inputs. But recent studies in the mouse show orientation selectivity in thalamic neurons, and address their potential source and possible roles in cortical computation.

Cristopher M. Niell

Open up a textbook on vision and you are likely to find a statement that the hallmark of visual cortex is 'orientation selectivity' — cells responding preferentially to edges or bars of light at a particular orientation. This is in contrast to cells in the retina and its relay to cortex, the lateral geniculate nucleus (LGN) of thalamus, where cells are generally considered to have circular receptive fields that are insensitive to orientation, and thus act more like spot detectors. Despite this dogma, however, it has long been known that cells in the retina perform a much broader array of visual processing, and indeed orientation selective neurons are present in the LGN of several species, suggesting these signals are conveyed to cortex (reviewed in [1]). Recently it has been shown that these selective cells are abundant in the mouse [2,3], constituting at least 10% of the population, making them more amenable to systematic study. Several

current studies, from Scholl *et al.* [4], Zhao *et al.* [5] and Lien and Scanziani [6], have started to address the potential impact of these selective cells on the cortex, but they leave open the question of what their actual significance is for cortical orientation selectivity.

Cortical Orientation and Direction Selectivity

Hubel and Wiesel [7] first demonstrated fifty years ago that, unlike neurons in the retina, cells in the primary visual cortex (V1) were best activated by edges or bars at a specific angle, generating a transformed representation of the visual world in terms of 'orientation selectivity'. They proposed a basic model, which has received significant confirmation, whereby these orientation-selective responses in simple cells could be built up from the untuned, circular receptive fields provided by LGN (Figure 1A).

A further subset of cortical cells shows a preference which direction

a bar or edge is moving, a property known as 'direction selectivity'. Direction-selective cells in the retina have also been known to exist for some time [8]; however, it was thought that these neurons projected to structures other than the LGN, and thus direction selectivity must also be computed anew in cortex.

Orientation Selectivity and Direction Selectivity in the Mouse LGN

The mouse has become an important model system for studying vision, largely because of the ability to genetically access defined cell types to assess their function and connectivity [9]. In fact, genetic markers for direction-selective retinal ganglion cells provided evidence that they do indeed project to the LGN [10]. Until recently, however, the mouse LGN had largely been unstudied (but see [11]), with work focusing on the retina and cortex.

Two recent studies [2,3] set out to specifically look for non-standard responses such as orientation selectivity in the LGN. Marshel *et al.* [2] developed a functional calcium-imaging preparation by removing the cortex to allow optical access to the superficial surface of the LGN, to record visually-evoked activity. Piscopo *et al.* [3] used silicon probes to perform multi-site electrophysiology, along with a battery of visual stimuli to probe for diverse response features,

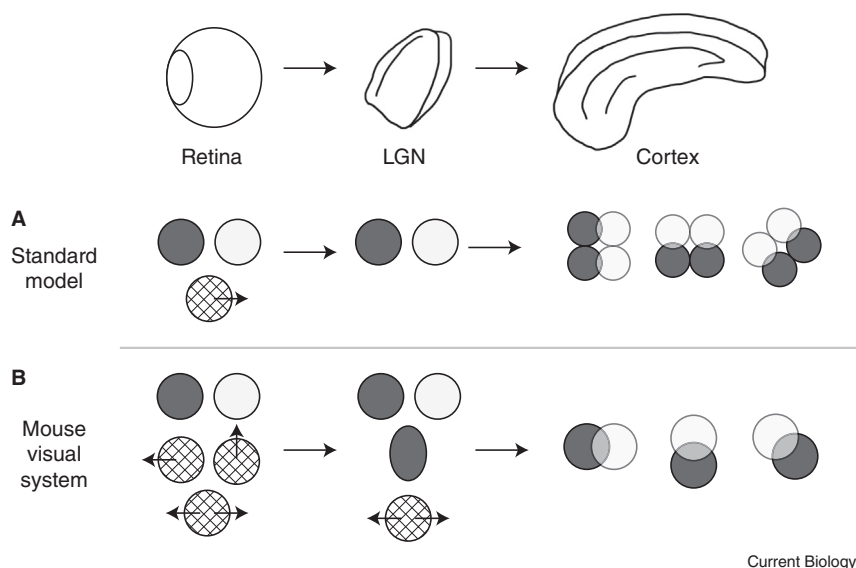


Figure 1. Organization of the early visual system.

(A) In the standard Hubel and Wiesel model, circular untuned receptive fields in retina are relayed to cortex by LGN, and summation of appropriately aligned untuned inputs creates selectivity for various orientations. While direction selective neurons are present in retina, they are not thought to project to cortex. (B) Current findings in the mouse reveal more diversity in the early pathways, including (but not limited to) both direction and orientation selectivity in the retina, and two types of orientation selective units in the LGN. However, thalamic inputs to layer 4 of cortex still appear to consist of appropriately aligned untuned units.

and a clustering algorithm to group neurons into classes with similar response profiles.

The majority of neurons in both studies resembled standard untuned center-surround LGN neurons. However, both groups found that ~10% of cells responded strongly to either one or both directions of motion of drifting bars, with most of the selective cells responding to both directions, indicating a preponderance of orientation selectivity. In both studies, selectivity was restricted to the four cardinal axes, similar to the direction-selective retinal ganglion cells that project to LGN, and in Marshel *et al.* [2] responses to motion along the horizontal direction were most prevalent. Furthermore, by reconstructing the recording sites from their multisite electrodes, Piscopo *et al.* [3] mapped the density of orientation-selective and direction-selective cells across the LGN, and found them located preferentially (though not exclusively) in the dorsolateral shell, where direction-selective retinal ganglion cell axons terminate.

These two studies [2,3] provided an initial characterization of this unexpected population, but raised

even more questions. Where does the LGN selectivity come from, and what does it do for cortical processing? Three more recent studies [4–6], which together describe recordings in retina, LGN, and cortex, in both mouse and cat, begin to provide answers.

Sources of LGN Orientation Selectivity

Although the LGN can act as a state-dependent gate, it is generally thought to relay signals from the retina without significant transformation of the spatial information represented [12]. Consistent with this, there is little recurrent connectivity among principal neurons of the LGN, and each LGN cell receives convergent input from only one to three retinal ganglion cells, limiting the amount of spatial transformation it can apply.

The LGN does receive feedback from V1, raising the possibility that orientation selectivity could be computed in cortex and then superimposed back on LGN. But when Zhao *et al.* [5] and Scholl *et al.* [4] tested this by inactivating cortex pharmacologically, they found no net change in the orientation selectivity of LGN, ruling out this possibility.

Zhao *et al.* [5] carefully examined the spatial receptive fields of orientation

selective LGN neurons, and showed that they fall into two types (Figure 1B). One third had a single elongated subregion, which could provide a basis for their selectivity, since the axis of elongation predicted the orientation of their selective response. As LGN cells can receive input from more than one retinal ganglion cell, these elongated receptive fields could result from the summation of two circular receptive fields that are slightly offset in location.

The other two thirds had circular receptive fields, which by themselves would not provide any spatial preference for one orientation. However, this is typical for direction-selective cells in the retina, suggesting a retinal source for these neurons' inputs. This still leaves open the question of how LGN could generate orientation selectivity (preference for either direction of motion of a single orientation). One obvious possibility is that LGN cells simply sum two direction-selective retinal ganglion cells with opposite preferred direction. In fact, Marshel *et al.* [2] originally proposed this based on their imaging data.

While Zhao *et al.* [5] neither prove nor disprove this possibility, they investigated the alternative hypothesis that orientation selectivity itself might be transmitted from the retina, by performing multi-electrode array recordings from isolated retinas *in vitro*. Strikingly, they found that a significant proportion of retinal ganglion cells (~20%) do indeed show orientation selectivity, which had not previously been described in mouse retina under normal conditions. This is in contrast to direction-selective retinal ganglion cells, which have been extensively studied, and for which genetic markers exist [13]. Because the major cell types of the mouse retina have been delineated based on cell morphology, it will be interesting to see to which morphological cell type this new response type corresponds.

Effects on Cortical Orientation Selectivity

Both Scholl *et al.* [4] and Zhao *et al.* [5] compared the level of orientation selectivity in the LGN to that found in the cortex, revealing a much greater selectivity in the spiking output of V1. Thus, it is not simply that V1 is identical to LGN in the mouse with no change in response properties. Furthermore, the

spatial receptive fields of mouse V1 neurons previously described [14] are quite different from those found in the present studies of the LGN.

However, Scholl *et al.* [4] found a surprising result when they examined the membrane potential tuning of cells in V1, which reflects the summed impact of all activity (both thalamic and cortical) coming into a cell, before the spike threshold mechanism. In contrast to cat, where the membrane potential tuning of cortical neurons is more selective than the average outputs from LGN, in the mouse they found a significant *decrease* in membrane potential tuning relative to the LGN selectivity. This indicated that, perhaps rather than building up greater selectivity from its inputs, the mouse cortex is discarding selectivity information. In fact, this leads to the provocative suggestion that the standard Hubel and Wiesel model of cortical orientation selectivity, via summation of properly aligned untuned LGN inputs, may not apply in the mouse.

Lien and Scanziani [6] independently addressed this possibility by directly examining the tuning of thalamic excitatory input to neurons in the thalamorecipient layer 4 of cortex. To isolate thalamic inputs, they transiently shut down cortex optogenetically by activating channelrhodopsin in a subclass of inhibitory interneurons. They then looked specifically at excitatory currents by performing whole-cell voltage clamp recordings.

The thalamic excitation Lien and Scanziani [6] recorded was consistent with predictions of the Hubel and Wiesel summation model. For most cells, they found both an ON and OFF subregion roughly the size of center-surround LGN receptive fields, which overlapped but were spatially offset (Figure 1B). As expected, the axis of this offset predicted the preferred orientation of the cell. By comparing the tuning of the periodic and time-averaged input in response to drifting gratings, they inferred that the orientation selective response observed was likely due to the summation of multiple inputs that themselves were untuned, as would be provided by standard center-surround LGN cells. Although their result does not completely rule out the possibility of some type of LGN orientation-selective cell providing tuning, such

hypothetical inputs do not match the described characteristics of LGN orientation-selective cells.

What Next?

If the LGN orientation-selective cells do not turn out to provide the primary source of orientation selectivity to layer 4 neurons, what role do they play in cortical processing? One proposal is that they may provide an initial selectivity bias early in development, which can then serve as a scaffold that is reinforced as further synaptic inputs are established [15].

Alternatively, the LGN orientation-selective neurons might provide a separate channel of input to a distinct subset of V1 neurons, much as magnocellular and parvocellular pathways provide parallel inputs to V1. It remains to be determined where the orientation-selective cells project to within cortex; indeed, koniocellular pathways in primate bypass layer 4 and target superficial layers directly [16]. If orientation-selective cells follow a similar logic, then their impact may be observed more directly outside of the primary thalamorecipient layer 4.

Of course, the ultimate test of their role will be causal manipulation, which is now possible in the mouse using optogenetic and pharmacogenetic methods. This approach will depend upon identifying genetic markers for this population in the LGN, or their potential inputs from the retina.

Is This Unique to Mouse?

Previous studies have documented LGN orientation and direction selectivity in several species, including rabbit [17], cat [18], and primate [19]. Scholl *et al.* [4] made a direct comparison of orientation selectivity in the cat and mouse LGN, and confirmed that orientation-selective cells are present in cat; however, there were four-fold fewer orientation-selective cells overall in cat.

If orientation-selective cells are present in other species, why are they rarely reported? This is likely due to their relatively low prevalence, which is exacerbated by their sub-localization within the LGN. In primate, they are often found in the koniocellular layers [20] (likely corresponding to the dorsolateral shell of mouse and deep layer C in cat), which are more difficult to target for recording.

Regardless of sampling, it seems clear that mouse simply has a higher prevalence of orientation-selective cells than other species. This may reflect the higher acuity of other species relative to mice — as other cell populations are expanded to sample visual space at higher density, the relative proportion of orientation-selective cells may decrease. It may also reflect the different uses for vision that have been specialized in the mouse.

However, the greater occurrence of this intriguing cell type in the mouse, along with the potential for genetic access, makes the mouse an excellent system for studying this pathway. It will be interesting to see if the recent interest in the diversity of LGN responses in mouse leads to an expanded view of the signals conveyed to the mammalian cortex across species.

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Calcium Signaling: Septins Organize the SOC Channel

Store-operated Ca^{2+} entry is a widely encountered mechanism for generating cytoplasmic Ca^{2+} signals. From a whole-genome screen in HeLa cells, a new study reveals that cytoskeletal septins play a critical role in this process, by organizing signaling structures in the plasma membrane.

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Store-operated Ca^{2+} entry (SOCE) is a process by which the level of Ca^{2+} stored in the endoplasmic reticulum (ER) regulates Ca^{2+} influx through plasma membrane Ca^{2+} channels [1]. SOCE was originally thought of as a mechanism to refill Ca^{2+} stores following their mobilization, for example, by inositol trisphosphate. However, it is becoming increasingly clear that SOCE serves as a primary source of signaling Ca^{2+} to which many downstream pathways of signal transduction are tightly linked [2,3]. Following the initial proposal of such a mechanism in 1986 [4], considerable effort focused on identifying the key molecular components of the pathway, notably the signal from the ER to the plasma membrane and the plasma membrane store-operated channel [5]. Ultimately, targeted and then whole-genome RNAi screens revealed the two major players: firstly, STIM proteins (in vertebrates, STIM1 and STIM2) in the ER serve as Ca^{2+} sensors; and, subsequently, activators of the plasma membrane store-operated channels, composed of Orai subunits (in mammals, Orai1, 2 and 3) [6,7].

Shortly after the discoveries of STIM and Orai, experiments overexpressing just two proteins, usually STIM1 and Orai1, produced huge intracellular Ca^{2+} signals, generating up to a 20-fold enhancement of the Ca^{2+} current

underlying SOCE [8]. This led to the preliminary conclusion that these two proteins might be the only obligatory constituents of the SOCE mechanism. However, the danger with overexpression experiments is that one generally does not have precise control or knowledge of the exact concentrations of the expressed proteins and cannot determine the efficiency of the process. In addition, there is the possibility of obligatory participants whose role is more catalytic than stoichiometric, such that the endogenous levels of these proteins are sufficient to permit STIM1–Orai1 interaction. A relevant example is found in a recent publication showing that knockdown of a specific phospholipase inhibits SOCE to the same extent as does knockdown of STIM1 or Orai1; however, when STIM1 and Orai1 are overexpressed, this phospholipase is no longer needed [9].

The laboratory that first identified Orai1 did so by screening a whole-genome *Drosophila* RNAi library and by monitoring the nuclear translocation of NFAT, a Ca^{2+} -activated transcription factor, as a readout of SOCE [10]. Reasoning that mammalian cells might regulate SOCE in more complex ways, a new study from this same laboratory now reports the screening of a whole-genome mammalian RNAi library in HeLa cells, again utilizing NFAT nuclear translocation as a marker of SOCE [11].

Sharma *et al.* [11] found that a strong inhibition of NFAT nuclear translocation occurred with a pool of siRNAs directed against septin 4, an siRNA pool that was later shown to reduce expression of septins 2, 4 and 5. With more specific duplexes, knockdown of all three septins was shown to be required for the reduction in NFAT translocation.

SOCE activation begins when low Ca^{2+} concentration in the endoplasmic reticulum leads to oligomerization of STIM1 and its accumulation at the plasma membrane in specific ER–plasma membrane junctions [12]. Subsequently, Orai1 is recruited to these junctions, leading to the formation of readily identifiable sites of concentration of both STIM1 and Orai1, termed puncta. There, SOCE ensues due to a direct interaction between STIM1 and Orai1 [7]. Orai1 can also be constitutively activated by expression of a portion of the soluble carboxyl terminus of STIM1. Sharma *et al.* [11] demonstrated that depletion of cellular septins inhibited activation of Orai1 by full-length STIM1 in response to Ca^{2+} store depletion, but did not inhibit constitutive activation of Orai1 by the carboxyl terminus of STIM1. Thus, neither the channel function of Orai1 nor its activation by STIM1 interaction appears to require septins. Rather it would seem that access of STIM1 to Orai1 must be impaired by septin depletion. Consistent with this interpretation, colocalization of STIM1 and Orai1 in puncta following Ca^{2+} store depletion was substantially reduced in septin-depleted cells [11]. When expressed alone (with only endogenous Orai1 present), the rate and extent of STIM1 movement to puncta was substantially diminished. Orai1 was also affected; in resting cells Orai1 is rather uniformly distributed in the plasma membrane, but when